

Ny Hasina Marolahy VAGNO

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PROFESSIONAL SUMMARY

Engineering student with 2+ years of experience in backend development, cloud infrastructure, and AI-driven geospatial systems.

TECHNICAL SKILLS

Programming: Java, Python, TypeScript

Backend: Spring Boot, REST APIs, Microservices, JUnit, Testcontainers

Cloud & DevOps: AWS (Lambda, SQS, EventBridge), Docker, CI/CD

AI/ML: YOLO, OpenCV, Scikit-learn, Pandas, GeoPandas

Data: PostgreSQL, GeoServer, OpenStreetMap

Frontend: Next.js, Firebase Auth

Tools: Git, Jira, SonarQube, QGIS, WebODM, Postman, Kaggle, Roboflow

PROFESSIONAL EXPERIENCE

Research Intern

Grupo de Redes de Computadores – Spain | Apr 2026 – Aug 2026

- Developed a UAV pipeline for citrus yield estimation from RGB aerial imagery
- Fine-tuned YOLOv11 models for fruit detection and canopy segmentation
- Processed geospatial orthophotos using WebODM, GDAL, and QGIS
- Built yield prediction models using spatial mapping and regression analysis

Software Developer

Numer Madagascar – Madagascar | Jul 2022 – Aug 2025

- Developed and deployed Spring Boot microservices handling 1,000+ requests/min
- Built serverless AWS pipelines (Lambda, SQS), improving scalability and reducing costs
- Fine-tuned YOLO models for high-resolution aerial imagery detection
- Automated geospatial vectorization pipelines using GeoPandas, reducing manual processing time
- Implemented unit and integration tests, achieving 80% code coverage

EDUCATION

Engineering Degree in Computer Science (Master's – Year 2)

ESIROI – Réunion Island | 2025 – Present

Bachelor in Computer Science

Haute École d'Informatique | 2021 – 2024

LANGUAGES

Malagasy (Native) | French (Bilingual) | English (TOEIC - B2)

INTERESTS

Basketball (Team Captain), Soccer, Music